

FTGP Group 11: Week 23 Sprint Report

EcoPassEU — Privacy-Aware Digital Product Passport DApp

Sprint Duration: 120 April 2026 – 30 April 2026

Report Date: 20 April 2026

University of Bristol

Last Sprint Review and Retrospective

Sprint Duration: 14 April 2026 – 20 April 2026

This section summarises the sprint review meeting held on 20 April 2026, marking the close of Sprint 1. The meeting was attended by all five team members in person: Luxiao Cao, Akshansh Rajora, Tianwei Yu, Detai Dong, and Fuyu Cao, who returned to in-person attendance this sprint having fulfilled her undergraduate mid-term dissertation defence obligations.

Completed Work

The following tasks were completed during Sprint 1, together with the primary contributors responsible for each item.

All sprint artefacts committed to GitHub

All members

All work produced during Sprint 1 was committed to the group's private GitHub repository under the organisation account shared with supervising teaching staff. This includes source code, data-processing scripts, smart contract code, and documentation. The repository is accessible at https://github.com/DeTaiDong/FTGP2425_Group1_11.

Paper writing: introduction and background

Luxiao Cao (lead)

Luxiao Cao led the academic writing effort this sprint, establishing the structural framework for the project paper and authoring both the introduction and background sections. The introduction situates EcoPassEU within the European Union's Ecodesign for Sustainable Products Regulation (ESPR) and articulates the motivation for a privacy-preserving, blockchain-based Digital Product Passport. The background section surveys relevant prior work spanning Life Cycle Assessment (LCA) methodology, decentralised identifier frameworks, and IPFS-based off-chain storage architectures. Both sections were written in Springer LNCS format, consistent with the group's agreed submission template.

Development environment support and application testing

Luxiao Cao

In addition to his writing responsibilities, Luxiao Cao assisted other team members in configuring and troubleshooting their local development environments, in particular resolving Node.js version conflicts encountered when running the Hardhat toolchain alongside the React frontend scaffold. He also conducted an initial round of functional testing on the deployed application, verifying wallet connection, QR code generation, and passport retrieval flows, and documented the results in a shared testing log committed to the repository.

Frontend: core implementation

Detai Dong (lead)

Detai Dong substantially completed the React frontend during Sprint 1. The delivered

implementation covers the three principal views specified in the sprint backlog: the passport display page, which renders normalised DPP data retrieved from IPFS; the QR code scanner and generator, integrated with the `qrcode.react` library; and the wallet connection component, implemented via Ethers.js and tested against the Polygon Amoy testnet. Detai Dong notes that a small number of supplementary features — including export-to-PDF and a comparative product view — were scoped but not yet delivered; these are planned for a subsequent sprint. The core functionality required for end-to-end integration testing is in place.

Smart contract and partial backend

Akshansh Rajora

Akshansh Rajora completed the ERC-1155 smart contract implementation, extending the skeleton from Sprint 0 to support full IPFS Content Identifier (CID) storage per token, on-chain event emission for passport issuance and updates, and access-controlled `mintBatch` and `updatePassportHash` functions. The contract was deployed to the Polygon Amoy testnet via Hardhat Ignition; the contract address and ABI were committed to the repository under `contracts/deployments/`. Akshansh Rajora additionally implemented supporting backend utility functions for CID retrieval and event indexing, which are consumed by the React frontend via Ethers.js.

Data processing and contract-ready field mapping

Fuyu Cao (async), Tianwei Yu

Fuyu Cao and Tianwei Yu jointly produced the data pipeline for Sprint 1. Working from the Open Beauty Facts CSV downloaded in Sprint 0, they applied a filtering strategy to produce a representative working subset of approximately 500 records, resolved UTF-8 encoding issues in the source file, standardised country-of-manufacture names to ISO 3166-1 alpha-2 codes, and imputed missing packaging fields using the documented fallback values agreed in Sprint 0. Critically, and in response to the schema-alignment issue identified during integration, the team ensured that all output JSON-LD field names and data types conform precisely to the smart contract ABI. Fuyu Cao produced the data gap log — a structured CSV recording each imputed field, its original value, the substituted value, and the justification — and submitted it to the repository ahead of the review meeting.

Changes and Incomplete Tasks

Frontend supplementary features — deferred to Sprint 2.

Detai Dong's additional frontend features (export-to-PDF, comparative product view) were not completed within the sprint. These are lower-priority enhancements relative to the core integration deliverables and have been carried forward to the Sprint 2 backlog.

Schema alignment issue — resolved mid-sprint.

An unanticipated mismatch was identified between the field names produced by the initial data pipeline and those expected by the smart contract ABI. This required a mid-sprint realignment between Fuyu Cao and Tianwei Yu (data pipeline) and Akshansh Rajora (smart contract), adding approximately half a day of unplanned work. The issue was resolved before the integration test and highlighted the need for an explicit shared schema document at the outset of future sprints.

Retrospective

Start Doing	Stop Doing	Keep Doing
Establish a formal mid-sprint check-in to review integration progress and surface blockers before the final two days of the sprint.	Treating integration testing as a final-day activity; begin smoke-testing component interfaces as soon as individual tasks reach a stable state.	Daily asynchronous standups, which continued to keep all members informed across time zones without requiring synchronous attendance.
Define and commit shared data-contract schemas (JSON-LD field names, types, and nullable flags) at the outset of each sprint, before implementation begins.	Leaving field naming conventions implicit; schema disagreements between the data pipeline and the smart contract cost time during integration.	Feature branches with pull-request review; the practice was adopted this sprint and reduced merge conflicts noticeably.
Include a brief written risk update in the repository at each sprint mid-point, flagging any emerging technical risks (e.g. IPFS bandwidth, testnet faucet availability).	Deferring risk documentation until the retrospective; risks identified late in the sprint (IPFS pinning limits) would benefit from earlier visibility.	Inclusive participation: all five team members contributed actively throughout the sprint, with clear task ownership and timely communication maintaining momentum across the group.

Next Sprint Planning

Sprint Duration: 21 April 2026 – 27 April 2026

The sprint planning meeting for Sprint 2 was held on 20 April 2026. The full team was present in person.

Product Owner

Tianwei Yu

Scrum Master

Detai Dong

Sprint Vision

By the end of Sprint 2, the team aims to have a fully integrated and evaluated prototype ready for academic reporting. The principal objectives are: completing the remaining frontend features; expanding and refining the academic paper with methodology, results, and evaluation sections; conducting a comprehensive evaluation covering hash integrity, gas costs, QR resolution latency, and frontend load time; and addressing any outstanding

technical debt from Sprint 1. The sprint will conclude with a near-final draft of the project paper and a stable, demonstrable prototype.

Sprint Backlog

The following tasks have been selected from the product backlog for Sprint 2. Estimates are given in story points (1 point $\approx$ half a day of focused work).

Task	Description	Assignee(s)	Points
Paper writing: methodology, results & evaluation	Draft the methodology, results, and evaluation sections of the project paper, incorporating gas cost data, load time measurements, and hash integrity results from Sprint 1.	Luxiao Cao (lead)	5
Application testing & bug fixes	Conduct systematic regression testing of the full application stack; document and resolve outstanding bugs identified during Sprint 1 integration testing.	Luxiao Cao	2
Frontend: supplementary features	Implement the deferred frontend features: export-to-PDF for passport records and a comparative product view for side-by-side DPP display.	Detai Dong (lead)	4
Smart contract: audit & optimisation	Review the deployed ERC-1155 contract for gas optimisation opportunities; add NatSpec documentation; update unit tests to achieve full branch coverage.	Akshansh Rajora	4
Data pipeline: extended dataset & ESPRESSO integration	Expand the working dataset to 1,000 records and integrate ESPRESSO V2 emission factors and water stress characterisation factors into the JSON-LD output.	Fuyu Cao, Tianwei Yu	5
Comprehensive evaluation	Run the full evaluation suite: hash integrity verification, QR code resolution latency (10 samples), gas cost per DPP issuance (mean and variance), and frontend load time. Produce tables and figures for the paper.	All members	5

Total story points this sprint: 25

Notes on Member Availability

Fuyu Cao has returned to in-person attendance from Sprint 2 onwards, having completed her undergraduate mid-term dissertation defence obligations. She will attend all sprint meetings in person alongside the rest of the team and will take ownership of the data pipeline expansion and ESPresso integration task.

All other team members are available full-time in Bristol for the sprint duration. Given the volume of academic writing required this sprint, Luxiao Cao has indicated that writing productivity may be sensitive to interruptions; the team has agreed to protect at least two uninterrupted working days for paper drafting. Detai Dong has confirmed that the deferred frontend features are scoped to four story points and are achievable within the sprint without impacting integration stability.

Anything Else to Share

The GitHub repository continues to serve as the single source of truth for all sprint artefacts. All code, documentation, data pipeline scripts, and test logs from Sprint 1 are committed and accessible to supervising teaching staff at https://github.com/DeTaiDong/FTGP2425_Group1_11.

The IPFS pinning bandwidth concern flagged in the Week 22 report remains under monitoring. Usage during Sprint 1, with a dataset subset of approximately 500 records, remained within the free-tier limits of the Pinata pinning service. Should the dataset grow significantly in Sprint 2 — as planned, to 1,000 records — the team will evaluate deployment of a self-hosted IPFS node on a university virtual machine as a contingency.

The end-to-end integration test conducted at the close of Sprint 1 was successful: all 500 records were ingested, pinned to IPFS, minted as ERC-1155 tokens on Polygon Amoy, and resolved correctly via QR code in the React frontend. Gas costs per DPP issuance and frontend load times were within acceptable parameters and will be reported formally in the Sprint 2 evaluation section of the paper.